Problem Set 4

Instructor solutions

Distributions, tails, and coupon collection

100 points + 10 optional challenge points • About 4 hours

Teaching note. Problems 3–4 deepen the coupon collector; Problems 6–7 develop tail areas and use both sides of a signed payoff. The challenge introduces unequal coupon probabilities. Award the indicated subpart points for correct reasoning, allowing follow-through credit after arithmetic errors. The tail proofs and challenge are guided enrichment.

Instructions. Show your reasoning and state each distribution's support and parameters. Geometric variables count trials including the first success; exponential parameters are rates. Write $H_r = \sum_{j=1}^{r} 1/j$ and $H_0 = 0$. Exact expressions suffice unless decimals are requested. A calculator or short program is allowed for numerical comparisons. Independent uniform choices of coupon types are assumed except in the challenge.

I. Discrete mechanisms

1. A count is not a wait

[12 points]

A box contains 20 distinguishable gadgets, of which 6 are defective.

(a) Choose one gadget uniformly. Let I indicate a defect. Give its law, mean, and variance. (2)

(b) Make four independent uniform draws *with replacement*. For the number B of defective gadgets drawn, give the law, $\mathbb{P}(B = 2)$, mean, and variance. (3)

(c) Choose four gadgets *without replacement*. For the defective count D, give its support, PMF, mean, and variance. Explain the variance comparison with (b). (4)

(d) Draw independently with replacement until the first defect. For the draw count G, give its PMF, $\mathbb{P}(G > 4)$, mean, and variance. Why is it not binomial? (3)

Solution

(a) $I \sim \text{Bernoulli}(3/10)$ on $\{0, 1\}$: $\mathbb{E}[I] = 3/10$, $\text{Var}(I) = 21/100$.

(b) $B \sim \text{Bin}(4, 3/10)$ on $\{0, \ldots, 4\}$. $\mathbb{P}(B = 2) = \binom{4}{2}(3/10)^2(7/10)^2 = 1323/5000$; $\mathbb{E}[B] = 6/5$, $\text{Var}(B) = 21/25$.

(c) D is hypergeometric with $N = 20, K = 6, m = 4$, supported on $\{0, \ldots, 4\}$, with $\mathbb{P}(D = k) = \binom{6}{k}\binom{14}{4-k}/\binom{20}{4}$. $\mathbb{E}[D] = 6/5$, $\text{Var}(D) = (21/25)(16/19) = 336/475$. Removing a defect lowers the chance of a defect next time: the draw indicators have negative covariance, reducing the variance.

(d) $G \sim \text{Geom}(3/10)$ on $\{1, 2, \ldots\}$, $\mathbb{P}(G = k) = (7/10)^{k-1}(3/10)$, $\mathbb{P}(G > 4) = (7/10)^4$, $\mathbb{E}[G] = 10/3$, $\text{Var}(G) = 70/9$. The number of trials is random; we record the first-success time, not the successes in a fixed number of trials.

2. Rare errors and a geometric tail

[10 points]

Each of 2000 independent transmitted bits is corrupted with probability 0.001. Let N count corrupted bits.

(a) Give the exact law of N and a suitable Poisson approximation. Compare the exact and approximate $\mathbb{P}(N \geq 2)$ to four decimals. (4)

(b) For $Q \sim \text{Pois}(\lambda)$, derive $\mathbb{E}[Q(Q-1)] = \lambda^2$ by shifting the PMF sum, and recover its variance. (3)

(c) In an unlimited stream with the same independent error probability, let G be the bit index of the first error. Use $\mathbb{E}[G] = \sum_{m=0}^{\infty} \mathbb{P}(G > m)$ to find its mean, and calculate $\mathbb{P}(G > s + t \mid G > s)$ for integers $s, t \geq 0$. (3)

Solution

(a) $N \sim \text{Bin}(2000, .001)$; the approximation is Pois(2). The probabilities are $1 - (.999)^{2000} - 2000(.001)(.999)^{1999} \approx .5941$ and $1 - 3e^{-2} \approx .5940$. The opportunities are numerous, independent, and rare, with mean count $2000(.001) = 2$.

(b) For $\lambda > 0$, $\sum_{k \geq 2} k(k-1)e^{-\lambda}\lambda^k/k! = \lambda^2 e^{-\lambda} \sum_{j \geq 0} \lambda^j/j! = \lambda^2$. Thus $\mathbb{E}[Q^2] = \lambda^2 + \lambda$ and $\text{Var}(Q) = \lambda$. At $\lambda = 0$ the variable is constantly zero.

(c) $\mathbb{P}(G > m) = .999^m$, so $\mathbb{E}[G] = 1/(1-.999) = 1000$ bits. The conditional probability is $.999^{s+t}/.999^s = .999^t$: past failures do not change the extra wait.

II. What the coupon-collector mean leaves out

3. The last few stickers [14 points]

An album has n equally likely sticker types. Let $T_{n,r}$ be the number of purchases needed to see r distinct types, $1 \leq r \leq n$, and let $T_n = T_{n,n}$.

(a) Define stages W_k, for $0 \leq k < n$, identify their laws, and justify their mutual independence from the conditional distribution after any completed history. (4)

(b) Derive $\mathbb{E}[T_{n,r}]$ and $\text{Var}(T_{n,r})$ as finite sums. Simplify the expectation with harmonic numbers. (4)

(c) For $n = 100$, compare the expected purchases for the first 90 types with the expected *additional* purchases for the last 10. Give one decimal place. Why do the last 10 take so long? (3)

(d) For $n = 6$, find $\mathbb{E}[T_6]$ and $\text{Var}(T_6)$ exactly. If exactly two faces are still missing, what is the expected remaining number of rolls? Why is T_6 not geometric? (3)

Solution

(a) W_k counts additional draws, including the next new type, after k distinct types have appeared. $W_k \sim \text{Geom}((n-k)/n)$. Given any particular completed history, future draws are independent and uniform; the chance of a new type stays $(n-k)/n$ until success. The conditional stage law depends only on k, not the previous stage lengths or labels. Successively multiplying these conditional PMFs gives the product of the marginal PMFs, proving stage independence. $W_0 = 1$.

(b) $T_{n,r} = \sum_{k=0}^{r-1} W_k$. Hence

$$\mathbb{E}[T_{n,r}] = n \sum_{j=n-r+1}^{n} \frac{1}{j} = n(H_n - H_{n-r}), \qquad \text{Var}(T_{n,r}) = \sum_{j=n-r+1}^{n} \left(\frac{n^2}{j^2} - \frac{n}{j} \right).$$

Linearity alone proves the mean; independence is needed for the variance.

(c) First 90: $100(H_{100} - H_{10}) \approx 225.8$. Last 10: $100H_{10} \approx 292.9$. With j types missing, a new one has probability $j/100$, so its expected wait is $100/j$. The very last type alone costs 100 purchases on average.

(d) $\mathbb{E}[T_6] = 6H_6 = 147/10$; $\text{Var}(T_6) = 36\sum_{j=1}^{6} j^{-2} - 6H_6 = 3899/100$. Two missing faces require $6/2 + 6/1 = 9$ more rolls on average. Completion cannot occur before six rolls, and finding a new face changes the success probability; T_6 is a sum of stages, not one geometric wait.

4. A deadline is not an expectation [14 points]

Let M_m count types missing after $m \geq 1$ purchases, with $n \geq 2$ types. Write $J_i = 1$ if type i is missing, $q = (1 - 1/n)^m$, and $b = (1 - 2/n)^m$.

(a) Compute $\mathbb{E}[M_m]$ and $\mathbb{E}[J_i J_j]$ for $i \neq j$. Deduce

$$\operatorname{Var}(M_m) = nq(1 - q) + n(n - 1)(b - q^2).$$

Are missing-type indicators independent? (4)

(b) Derive the exact completion CDF by inclusion–exclusion: $\mathbb{P}(T_n \leq m) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (1 - j/n)^m$. (3)

(c) For six types, compute $\mathbb{P}(T_6 \leq 15)$ and $\mathbb{E}[M_{15}]$ to four decimals. Find the smallest integer m with $\mathbb{P}(T_6 \leq m) \geq .95$; report probabilities at $m - 1$ and m. (4)

(d) Prove $\mathbb{P}(T_n > m) \leq n e^{-m/n}$. For 100 types choose an integer m guaranteeing at least 95% completion by this bound. Does the bound establish the smallest possible deadline? (3)

Solution

(a) $M_m = \sum_i J_i$, so $\mathbb{E}[M_m] = nq$. Two specified types are both absent with probability b, since each draw avoids them with probability $1 - 2/n$. Expanding the variance of a sum gives the stated formula. For $m \geq 1$, $b < q^2$ because $1 - 2/n < (1 - 1/n)^2$. Thus the distinct indicators have negative covariance and are dependent. (At $m = 0$ all indicators are constant.)

(b) Completion is the complement of the union of missing-type events. A specified set of j types is absent with probability $(1 - j/n)^m$; there are $\binom{n}{j}$ such sets. Inclusion–exclusion gives the result. At $m = 0$ the CDF is zero; handle it directly to avoid 0^0 conventions.

(c) $\mathbb{P}(T_6 \leq 15) \approx .6442$, $\mathbb{E}[M_{15}] = 6(5/6)^{15} \approx .3894$. The smallest deadline is $m = 27$, since $\mathbb{P}(T_6 \leq 26) \approx .9480$ and $\mathbb{P}(T_6 \leq 27) \approx .9566$. The CDF is nondecreasing. Fifteen rolls are close to the mean 14.7, yet completion is far from certain.

(d) The union bound gives $\mathbb{P}(T_n > m) \leq n(1 - 1/n)^m \leq n e^{-m/n}$. Take $m = \lceil 100(\log 100 + \log 20) \rceil = 761$. This is a sufficient deadline; the bound does not determine the exact minimum. At the chosen m, its failure bound is about .04955.

III. Densities and tail areas

5. A density that exceeds one [12 points]

Suppose $f_X(x) = cx^2$ for $0 \leq x \leq 2$, and zero elsewhere.

(a) Normalize the density and write its full piecewise CDF. (3)

(b) Find $\mathbb{P}(1/2 < X \leq 1)$, $\mathbb{E}[X]$, and $\operatorname{Var}(X)$. (3)

(c) Find the median and $\mathbb{P}(X = 1)$. Explain why $f_X(2) > 1$ is allowed. (3)

(d) Let $Y = (X/2)^3$. Find $F_Y(y)$ by rewriting $\{Y \leq y\}$ as an event about X, and identify the distribution of Y. (3)

Solution

(a) $1 = c \int_0^2 x^2 \, dx = 8c/3$, so $c = 3/8$. $F_X(x) = 0$ for $x < 0$, $x^3/8$ for $0 \leq x \leq 2$, and 1 for $x > 2$.

(b) $\mathbb{P}(1/2 < X \leq 1) = 7/64$; $\mathbb{E}[X] = \int_0^2 (3/8)x^3 \, dx = 3/2$; $\mathbb{E}[X^2] = \int_0^2 (3/8)x^4 \, dx = 12/5$. Thus $\operatorname{Var}(X) = 12/5 - 9/4 = 3/20$.

Solution (continued)

(c) The median is $2 \cdot 2^{-1/3}$, solving $F_X(x) = 1/2$. $\mathbb{P}(X = 1) = 0$. Density is probability per unit length; only its integrated area must be at most one on an interval. Here $f_X(2) = 3/2$.

(d) For $0 \le y \le 1$, $F_Y(y) = \mathbb{P}(X \le 2y^{1/3}) = (2y^{1/3})^3/8 = y$. The CDF is zero below 0 and one above 1, so $Y \sim \text{Uniform}(0, 1)$.

6. Tail areas and the suspiciously generous prize [12 points]

For a real variable Z, write $F(z) = \mathbb{P}(Z \le z)$ and assume $\mathbb{E}[|Z|] < \infty$ in (a)–(b).

(a) For $Z_+ = \max(Z, 0)$ and $Z_- = \max(-Z, 0)$, use $y = \int_0^\infty \mathbf{1}_{\{y>t\}}\, dt$ for $y \ge 0$ to show

$$\mathbb{E}[Z] = \int_0^\infty (1 - F(t))\, dt - \int_{-\infty}^0 F(t)\, dt.$$

You may interchange expectation and nonnegative integrals. Explain why using $\mathbb{P}(Z < t)$ instead of $\mathbb{P}(Z \le t)$ does not change an integral. (3)

(b) What does the *sum* of these two nonnegative tail areas compute? Check both identities for the constant payoff $Z = -2$. (3)

(c) A carnival advertises a prize R (in dollars) with CDF zero for $x < 1$ and $F_R(x) = 1 - 1/x$ for $x \ge 1$. Show that this is a valid continuous law, find its density, and use a tail integral to find its mean. Is there a finite fair entry fee for the uncapped game? (3)

(d) The small print caps the payout at $B > 1$ dollars: $C = \min(R, B)$. Use its tail to find the fair entry fee. What probability mass is at B? Evaluate the fee for $B = 100$. (3)

Solution

(a) Taking expectations in the indicator identity gives $\mathbb{E}[Z_+] = \int_0^\infty \mathbb{P}(Z > t)\, dt$ and $\mathbb{E}[Z_-] = \int_0^\infty \mathbb{P}(Z < -t)\, dt$. A distribution has at most countably many atoms: the jump locations can change the integrand only at a set of length zero. Substitute $u = -t$ in the second integral and use $Z = Z_+ - Z_-$. The integrability assumption makes both areas finite.

(b) The sum equals $\mathbb{E}[Z_+ + Z_-] = \mathbb{E}[|Z|]$. For $Z = -2$, the negative-side area is 2 and the positive-side area is 0; the signed mean is -2, while the sum is 2.

(c) The CDF is continuous and nondecreasing, with limits zero and one. Its density is x^{-2} on $x > 1$, zero below 1, and has total area one. The tail equals 1 on $[0, 1)$ and $1/t$ for $t \ge 1$. Thus $\mathbb{E}[R] = 1 + \int_1^\infty dt/t = +\infty$. There is no finite fee making expected net payoff zero. This is an infinite positive mean, not an undefined difference of two infinite areas.

(d) $\mathbb{P}(C > t) = 1$ for $0 \le t < 1$, $1/t$ for $1 \le t < B$, and zero for $t \ge B$. Therefore $\mathbb{E}[C] = 1 + \log B$ dollars, the fair entry fee. $\mathbb{P}(C = B) = \mathbb{P}(R \ge B) = 1/B$: the capped law is a mixture of a density below B and an atom at B. At $B = 100$, the fair fee is $1 + \log 100 \approx \$5.61$, despite the uncapped infinite mean.

IV. Continuous mechanisms and a signed payoff

7. The auction for the last slice [12 points]

At a pizza party, $n \geq 1$ hungry mathematicians independently submit bids $U_1, \ldots, U_n$, each uniform on $[0, 1]$ dollars. The seller receives the *highest bid* $M = \max_i U_i$, even if it is small. The slice cost c dollars, where $0 < c < 1$, so the seller's net payoff is $Z = M - c$.

(a) Find the full CDF of M, and then of Z, using independence. (3)

(b) Compute separately the negative-tail area $L = \int_{-\infty}^{0} F_Z(z)\,dz$ and positive-tail area $G = \int_0^{\infty}(1 - F_Z(z))\,dz$. Use $G - L$ to find the mean profit *without differentiating the CDF*. (4)

(c) Find the loss probability and $\mathbb{E}[|Z|]$. Explain why the latter need not equal the absolute value of the mean profit. (3)

(d) For $n = 3, c = 3/4$, calculate the mean profit, loss probability, and mean absolute payoff. What does this say about calling a game "fair"? (2)

Solution

(a) On $0 \leq x \leq 1$, $F_M(x) = \prod_i \mathbb{P}(U_i \leq x) = x^n$; it is zero below 0 and one above 1. $F_Z(z) = 0$ below $-c$, $(z + c)^n$ on $[-c, 1 - c]$, and 1 above $1 - c$.

(b) Substitute $u = z + c$:

$$L = \int_{-c}^{0} (z + c)^n\,dz = \frac{c^{n+1}}{n+1}, \qquad G = \int_0^{1-c} [1 - (z + c)^n]\,dz = \frac{n}{n+1} - c + \frac{c^{n+1}}{n+1}.$$

Hence $\mathbb{E}[Z] = n/(n + 1) - c$; fairness requires $c = n/(n + 1)$. L is the expected size of losses, counting zero on profitable outcomes; G is the expected size of gains, counting zero on losing outcomes. Neither is a conditional average given a loss or gain.

(c) $\mathbb{P}(Z < 0) = c^n$ and $\mathbb{E}[|Z|] = G + L = n/(n + 1) - c + 2c^{n+1}/(n + 1)$. The absolute value before averaging removes cancellations between losses and gains. At a fair price, the signed mean is zero but the absolute mean is positive.

(d) Mean profit: 0. Loss probability: $(3/4)^3 = 27/64$. Mean absolute payoff: $2(3/4)^4/4 = 81/512$ dollars. Fairness is zero expected net payoff; it need not mean a one-half chance of losing or an absence of risk.

8. A clock, a bus, and a normal measurement [14 points]

(a) A service has independent Poisson counts on disjoint intervals, with mean $3t$ requests in t minutes. Let W be the time to the next request. Derive its survival function from a zero-count probability; find its mean, variance, and median. (4)

(b) Compute $\mathbb{P}(W > 5/3 \mid W > 1)$ and compare it with $\mathbb{P}(W > 2/3)$. A bus wait V is instead uniform on $[0, 2]$ minutes. Compute $\mathbb{P}(V > 5/3 \mid V > 1)$ and $\mathbb{P}(V > 2/3)$; compare. (4)

(c) The duration X of a task is modeled as normal with mean 10 and variance 9 minutes squared. Express $\mathbb{P}(8 < X < 13)$ and $\mathbb{P}(X > 13)$ in terms of the standard normal CDF Φ. (3)

(d) What density-height or endpoint mistake appears in "$\mathbb{P}(X = 10) = 1/(3\sqrt{2\pi})$"? For $Y = 2X - 5$, give the distribution, mean, and variance. (3)

Solution

(a) $\mathbb{P}(W > t) = \mathbb{P}(\text{Pois}(3t) = 0) = e^{-3t}$ for $t \geq 0$. Thus $W \sim \text{Exp}(3)$, with mean 1/3 minute, variance 1/9 minutes squared, and median $(\log 2)/3$ minute. For example, its mean is the tail integral $\int_0^\infty e^{-3t}\,dt$.

(b) For the exponential, $e^{-5}/e^{-3} = e^{-2} = \mathbb{P}(W > 2/3)$. For the uniform, $(1/6)/(1/2) = 1/3$, whereas $\mathbb{P}(V > 2/3) = 2/3$. Only the exponential is memoryless.

(c) The standard deviation is 3. The probabilities are $\Phi(1) - \Phi(-2/3)$ and $1 - \Phi(1)$.

(d) A density height is not a point probability: $\mathbb{P}(X = 10) = 0$. $Y \sim \text{Normal}(15, 36)$, where the second parameter is variance. Its mean is $2(10) - 5 = 15$ and variance is $2^2(9) = 36$.

Optional challenge: the rare sticker

[10 extra points]

Now type i has probability $p_i > 0$, with $\sum_{i=1}^n p_i = 1$. Purchases are still independent. Let T count draws until all types appear. Write $p_A = \sum_{i \in A} p_i$.

(a) Explain why the next-new-type probability after k types have appeared can depend on *which* types they were. Derive $\mathbb{P}(T > m) = \sum_{\varnothing \neq A \subseteq \{1,\ldots,n\}} (-1)^{|A|+1}(1 - p_A)^m$ for $m \geq 1$. (3)

(b) Use the discrete tail sum to obtain $\mathbb{E}[T] = \sum_{\varnothing \neq A} (-1)^{|A|+1}/p_A$. For three types with probabilities $1/2, 1/3, 1/6$, evaluate it and compare with three equally likely types. (4)

(c) For two types with probabilities $p, 1 - p$, show the mean is $1/p + 1/(1 - p) - 1$ and is minimized uniquely at $p = 1/2$. (3)

Solution

(a) After seeing the set S, the new-type probability is $1 - \sum_{i \in S} p_i$, which need not be determined by $|S|$. A specified nonempty set A is missing after m draws with probability $(1 - p_A)^m$; inclusion–exclusion for the union of these missing events gives the tail. Uniform geometric-stage parameters no longer apply.

(b) The $m = 0$ tail is 1. The alternating subset coefficients also sum to 1, so we can include that term with the convention that each set is missing before any draw (including the full set). There are finitely many subsets and every $p_A > 0$, so their convergent geometric series may be summed separately: $\sum_{m \geq 0} (1 - p_A)^m = 1/p_A$. For three types,

$$\mathbb{E}[T] = 2 + 3 + 6 - \left(\frac{6}{5} + \frac{3}{2} + 2\right) + 1 = \frac{73}{10} = 7.3,$$

compared with $3H_3 = 11/2 = 5.5$ for uniform types.

(c) $\mathbb{E}[T] = 1/[p(1 - p)] - 1 \geq 4 - 1 = 3$, since $p(1 - p) = 1/4 - (p - 1/2)^2 \leq 1/4$. Equality holds exactly at $p = 1/2$.